

# Project 2

# CS 261

Save your program for this project as `<emailID>_project2.py` where `<emailID>` is the part of your Hampden-Sydney e-mail address before the `@` symbol. When you are finished, e-mail your program to `blins@hsc.edu`. Your solution is due by noon on Friday, October 2.

## Caesar Cipher

The Caesar cipher is one of the oldest and simplest methods for encrypting text. You pick a shift amount and then you shift every letter in the message by that amount. For example, if the shift is 2, then a's become c's, b's become d's and so on, until y and z wrap around to become a and b, respectively. A message like:

about to get hazy

would become:

cdqvv vq igv jcba.

For this project, you will write three functions related to the Caesar cipher. You may also include any additional functions that are helpful to construct the three functions described below. You should not import any libraries for this project.

1. Write a function `letter2number` that converts lower case letters (a, b, c, ..., z) to the numbers (0, 1, 2, ..., 25). For example `letter2number("c")` should return 2.
2. Write a function `number2letter` that converts a number to the corresponding (lower case) letter of the alphabet. For example, `number2letter(1)` should return "b".
3. Write a function called `caesar_shift(message, shift)` that applies a Caesar shift to any string. You can assume that the message will not have any upper case letters, and you only need to shift letters, not any other symbols.
4. The Caesar cipher is not a good way to encrypt a message because it is very easy to decrypt. In fact, you can often figure out the shift by finding the most common letter in the encrypted text. Since "e" is the most common letter in most English texts, there is a good chance that the shift is just the amount needed for "e" to become the most common letter in your encrypted message. Write a function called `letter_frequency(message)` that returns a list. The value of the output list at index number *i* should be the number of times the *i*-th letter of the alphabet occurs in the message. So the first entry of the output is the number of a's, then b's, and so on.
5. In order to find the letter that occurs the most in the message, we need a way to search through the output of the letter frequency list and find the index with the largest entry. Write a function called `argmax(list)` that will search a list of numbers and return the *index* of the largest entry in the list. If multiple entries in the list are tied for the largest value, then return the index of the first one that is equal to the maximum.

## Hints

1. To write the functions `letter2number` and `number2letter`, it would help to use the following string.

```
alphabet = "abcdefghijklmnopqrstuvwxyz"
```

Recall that you can use the string method `.find()` to find the index of any letter in the `alphabet` string.

2. You will definitely need to use modular arithmetic at some point during this project. It might help if your `number2letter` function uses modular arithmetic to handle numbers bigger than 25. For example, it might be nice if 26 wraps around to 0 and returns the letter "a".
3. Don't forget methods like `.count()` when working with lists and strings.
4. For the `argmax()` function, you'll need to loop over every value in the input list. Make sure to use an accumulator variable to keep track of which index corresponds to the maximum value you have seen so far.